other interfacing devices will move to Ethernet. Basic sensors for pressure and temperature will continue to use Fieldbus, but the Fieldbus network will be connected to a gateway with Ethernet as the backbone (Fig. 5). The reason why some sensors will move to Ethernet and others will not is mainly a trade-off between such criteria as the required data rate, Ethernet connectivity cost and sensor real estate.

To succeed in replacing Fieldbus on complex sensors and gateways, all the available Fieldbus features must also be available in industrial Ethernet. The first and most important feature of Fieldbus is its capability to deliver power over the network cable. PoE on industrial Ethernet addresses this requirement.

Another feature of Fieldbus is that it can be wired in a bus or tree topology instead of a star. By adding additional data link or application stacks, some of the industrial Ethernet protocols support line and tree network topologies. With this feature, the Ethernet network can be configured as a star, bus or tree.

How can PoE devices be combined to support a bus configuration? Adding a single-port PSE device combined with a PD in every node allows power to be received and forwarded in every node. Of course, the length of the line and number of devices on the line are limited by the maximum power injected at the beginning of the line.

PoE adds robustness, security and redundancy

Currently, many PLCs, I/O modules and drives have industrial Ethernet connections. However, their power requirements do not allow them to be powered over the Ethernet cable. Digital output modules can drive up to 500 mA per port at 24V, resulting in 12W per output. Instead of bringing power to the complete module, it makes sense to only supply the communication and control hardware

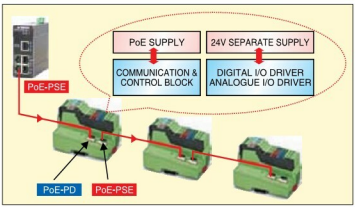


Fig. 6: PoE ensures communication between modules if the power line drops out

block via PoE. The rest of the module is powered over the 24V industrial power line.

PoE ensures communication between the modules if the 24V or AC mains power drops out (Fig. 6). The control part of the module is always available and ensures correct module shutdown. In this way, PoE guarantees a secure operation mode and secured storage of process data on the industrial system.

While the current IEEE 802.3af standard limits the power level to 13W, the new IEEE 802.3at standard brings the power level to 30W. This evolution makes it possible to bring PoE to a higher level of industrial devices. With more power available, high-end devices such as sensors, HMI, industrial cameras and RFID access points will be supported. Also, a longer bus or tree of PLC, I/O or drive module communication blocks can be supplied.

Fig. 6 shows a combination of PSE devices and PDs supporting bus and tree network topologies. With more power available, PoE could support longer buses or trees.

PoE, as defined in the IEEE 802.3 standard, is very closely linked with the physical layer and, in particular, the cable media. Typically, for the 10/100BASE-T physical media, Cat3-Cat5e cables are used. As mentioned earlier, all standards of industrial Ethernet use the same physical layer. Theoretically, PoE can be implemented on all of these standards.

The standard is to use Cat5 cable for IEEE 802.3af. Cat5e or higher has been proposed for IEEE 802.3at. The main argument for defining a different cable for higher power is the cable resistance. Cat5e and Cat6 have a 100m loop resistance of 25W compared to 40W for standard Cat3-Cat5 cable. This requirement reduces the voltage drop and power dissipation in the cable. Industrial Ethernet uses Cat5-6 cables and therefore does not add any restriction to the PoE technology.

Industrial Ethernet currently uses RJ-45 and M12 connectors (besides optical connectors). The RJ-45 has four pairs connected, while M12 connectors have only two pairs. Both the PoE standards—IEEE 802.3af and IEEE 802.3at—will use only two pairs for distributing the power. Therefore the different connectors will not restrict adoption of PoE.

Ready to power the network

IP convergence will drive industrial Ethernet from the enterprise to the factory floor. And PoE is ready to be introduced across the complete industrial Ethernet. PoE+ removes the power restriction on industrial Ethernet controllers and network topologies. In the future, industrial Ethernet will migrate farther down to the Fieldbus device level.

PoE can only strengthen industrial Ethernet, positioning it to become the leading industrial network standard. ●

Concluded

Satya is a key member at Shresta eTechnologies, Bangalore, engaged in the design and development of software for embedded solutions, focused on communication, consumer, power and industrial automation segments